BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Hyderabad Campus

Second Semester 2021-2022
ECE F341/EEE F341/INSTR F341: Analog Electronics

Comprehensive Exam (Closed Book)

Part-B

Time-: 14:00 – 17:00 **Max Marks-:45** **Date: 12-05-2022**

Instructions: - There are three questions and marks for each question are indicated on the right. Read the questions carefully and write the answers neatly. Answer all the questions.

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1. (a) Evaluate the output voltage and node voltage values for the following op-amp circuit shown in Figure 1. The resistance values are given as R_1 = 2 Ω, R_2 = 3 Ω, R_3 = 4 Ω, R_4 = R_5 = R_6 = 10 Ω, R_F =500 Ω. The voltage values V_A=10 V, V_B=10 V and V_C =10 V should be considered for this circuit. Clearly write all steps to evaluate the node voltages and output voltage values for the op-amp circuit. **[8 Marks]**

   (b) Solve the following differential equation using op-amp circuit. You must have to use 2 integrators, and 2 adders for the implementation. Clearly write the mathematical expression for each op-amp output.

$$\frac{d^2v}{dt^2} + p\frac{dv}{dt} + qv = v1$$

   Where p and q are constants. similarly, v and $v1$ are the voltage values used in the op-amp stages. You can use resister and capacitor values in the op-amp stages as per your own choice. **[7 Marks]**

2. (a) Design a second order VCVS low pass-filter with pass band gain of 20 dB and cut-off frequency of 10 kHz. Use the capacitance values of the design to be 1 µF only. Draw the circuit with appropriate values. **[7 Marks]**

   (b) The Figure 2 shows a circuit of a simple digital thermometer. Given R = 100 Ω, R_1 = 10 kΩ and R_f = 30 kΩ and supply voltage = 15 V. R_1 is a RTD with a positive temperature coefficient 0.38 Ω/^{0}C. At 0^0C, the resistance of RTD = R = 100 Ω. The reference voltage for the 8-bit ADC is 12.8 V. Let the stimulus applied to convert and status inputs of ADC be a pulse of 12.5 µs starting at 12.5 µs. Find $V_{0,}$ the actual analog and digital temperature output at 10 ^{0}C and 20 ^{0}C. **[8 Marks]**

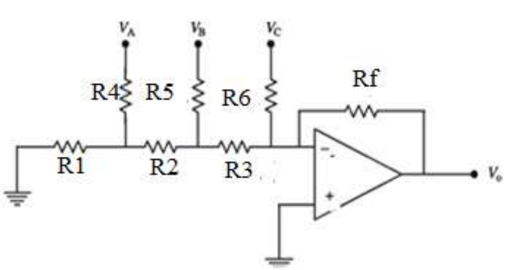


Figure 1: Circuit for Qn 1(a)

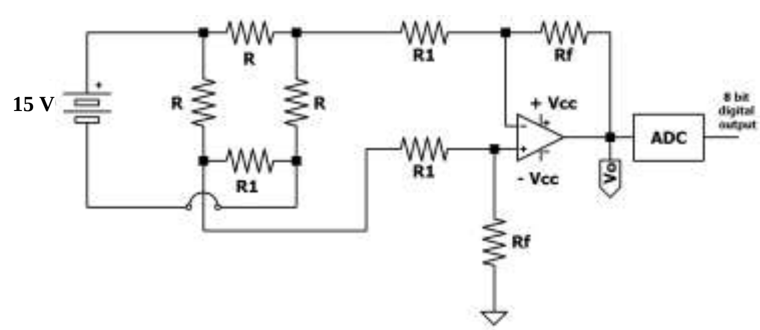


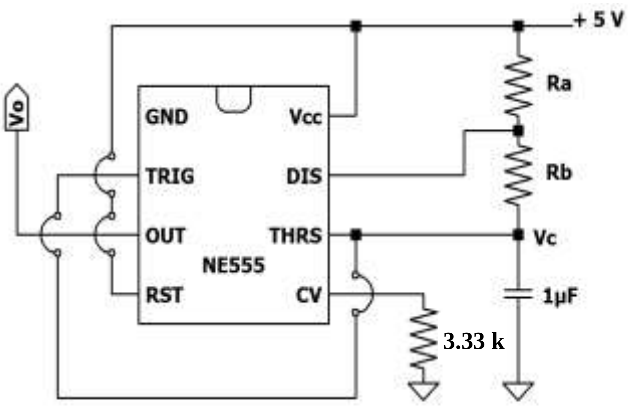
Figure 2: Circuit for Qn 2(b)

3. (a) A schmitt trigger with hysteresis width V_H = 2 V and lower threshold level V_{LT} = -1.5 V converts a 1 kHz sine wave of amplitude 5 V_{p-p} into a square wave. Calculate the value of V_{UT}. Also, calculate the time duration of the negeative and postive portioin of the output waveform. Draw the input and output waveform in the same scale. **[5Marks]**

(b) The Figure 3 shows a 555 timer in astable operation.

i. Find the threshold levels of upper and lower comparators of the timer circuit
ii. Find the time period for which the output is high (T_H)
iii. Find the time period for which the output is low (T_L)
iv. Find the relationship between R_A and R_B for which the two time intervals are equal.
v. What is the period of the output when the two time intervals are equal?

[10 Marks]



DIS – Discharge
TRIG – Trigger
THRS – Threshold
CV – Control Voltage
RST – Reset
GND - Ground

Figure 3: Circuit for Qn 3(b)